

Using R for reproducible data management and analyses in Forest Sciences

Post-graduate program in Forest Sciences / UnB

Discipline – Transfer of forestry information (Code: EFL360457)

Credits - 4

Workload – 60h (about 42h presential)

Schedule - Second semester of 2026 in a semi condensed way (8th to 11 of September and 5th to 9th of October, see program below)

Focus areas - Nature conservation, Forest management and Technology and use of forest resources

Level- Master and PhD

Professor – Géraldine Derroire, geraldine.derroire@cirad.fr

Learning outcomes

At the end of this module, the students will be able to:

- Define the concept of reproducible research and cite the main tools that it requires
- Understand the basis of R and R studio (interface, packages, basic functions...)
- Import, manipulate and export data in R
- Apply good practices in data management
- Organise their work in a R studio project
- Use the main packages of the Tidyverse for tidy analyses
- Choose the appropriate graphics to explore their data and create them using ggplot
- Implement common statistical tests in R
- Understand and implement more advanced programming techniques (loops, functions...)
- Use the tool Quarto to produce a report or a presentation with text and code
- Use git and GitHub for version control

Targeted audience

This module is open to students from the three focus areas, level Master and PhD.

No previous experience with R is required, as the course will start with the basics. We will progress to more advanced skills progressively, so the course can also be beneficial for more experienced students.

Teaching methods

The teaching methods will emphasise on interactivity and practical application. For this, each class will contain a mix of formal teaching, practical exercises, and discussion.

The module will be taught mostly in English: presentations will be in English, but questions and students' interventions can be made in either English or Portuguese. Course material will be provided in English but some suggested additional resources will be in Portuguese. This pedagogical choice aims at encouraging the students to work and communicate in English, in a friendly and inclusive way.

The sessions of the first week will set the foundations of using R, while the sessions of the second week will present more advance skills. Students are therefore encouraged to work independently or in groups between the two weeks. Corrected exercises will be provided to this aim.

Program

Please note that the starting time of the classes is not the same for all sessions.

	Dates	Time	Topic
1	Tuesday 8 th September	9-12 h	Introduction of the course, the professor and the students Principles of reproducible science Getting started with R
2	Tuesday 8 th September	13-16h	Getting started with R
3	Wednesday 9 th September	9-12 h	Working with data
4	Wednesday 9 th September	13-16h	Introduction to the tidyverse
5	Thursday 10 th September	9-12 h	Introduction to the tidyverse
6	Thursday 10 th September	13-16h	Data visualisation with ggplot
7	Friday 11 th September	9-12 h	Literate programming using quarto
8	Friday 11 th September	13-16h	Version control with git and GitHub
9	Monday 5 th October	8-12h	Statistics with R: common statistical tests
10	Tuesday 6 th October	8-12h	Statistics with R: linear models
11	Wednesday 7 th October	8-12h	Statistics with R: multivariate analyses
12	Thursday 8 th October	9-12h	Going further with R
13	Friday 9 th October	9-12h	Going further with R

Assessment

Student's learning will be evaluated through a final home assessment using the skills covered in class. **The exam instructions will be released on October 9th and the completed assignment must be submitted by November 9th.**

Practical aspects

Students will be encouraged to practice on their own laptop, in order to have all the programs installed for their personal work during and after the module. Should this be a problem, please inform the professor no later than the 25th of August so that an alternative solution can be looked for.

Students are asked to have installed the required programs before the first session. For this, a detailed tutorial is available here:

https://geraldinederroire.github.io/Course_R_Forest_Sciences/0_installing_enviro

Recommended bibliography

Additional resources will be proposed during the module.

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science* (2nd éd.). O'Reilly Media. <http://r4ds.had.co.nz/>

Wickham, H. (2017). *ggplot2 : Elegant graphics for data analysis* (2nd éd.). Springer. <https://ggplot2-book.org/>

Navarro, D. (2019) *Learning statistics with R: A tutorial for psychology students and other beginners*. <https://learningstatisticswithr.com/book/>

Damiani, A. Milz, B., Lente, C., Falbel, D., Correa, F., Trecenti, J., Luduvise, N., Lacerda, T., Amorim, W. (2022) *Ciência de Dados em R*. <https://livro.curso-r.com/index.html>